Chemistry • Hazards, PPE and emergency response

Laboratory and Practical Skills

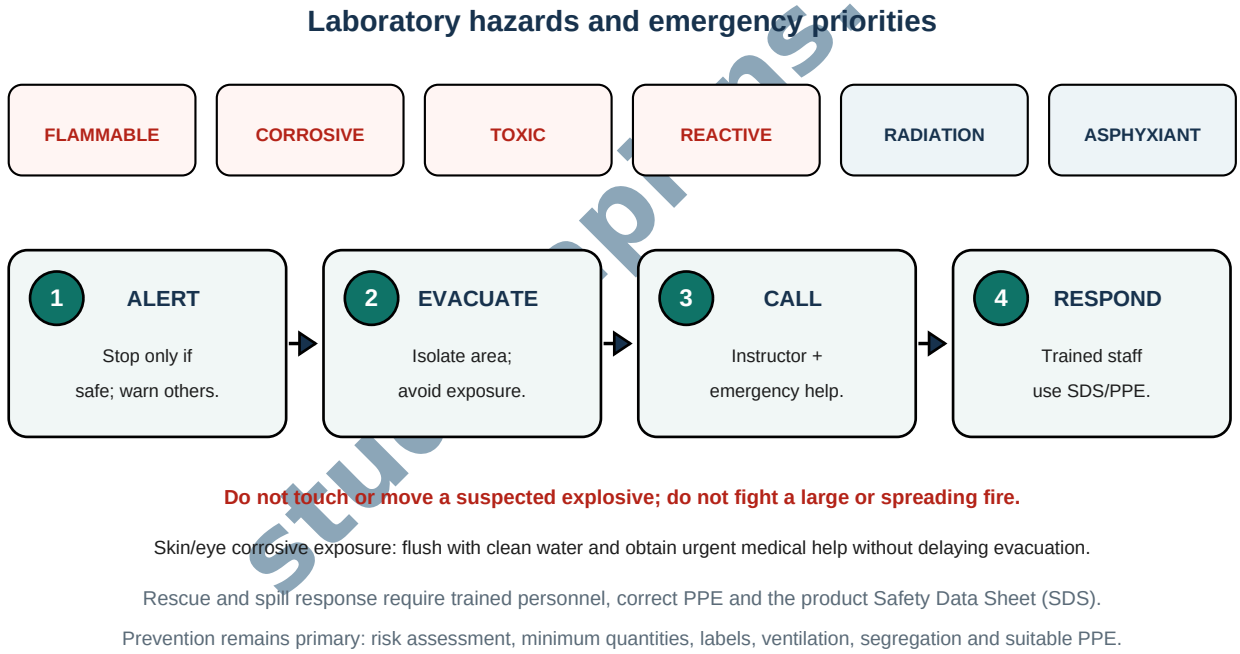
Official exercise • Complete solved answers • Modern safety guidance

Chapter 13 • Complete official exercise • 8 MCQs + 8 short + 5 constructed + 5 descriptive + 1 investigative

What this chapter is about

• **Hazard classes:** flammable/explosive, corrosive, toxic, reactive, radiation and asphyxiation hazards.
• **Prevention comes first:** risk assessment, substitution, minimum quantity, labels/SDS, segregation, ventilation, supervision and suitable PPE.
• **Know before starting:** exits, eyewash, safety shower, extinguisher, gas/electrical isolation, spill procedure and assembly point.
• **Emergencies:** alert, evacuate/isolate and obtain trained help; do not attempt unprotected rescue or fight a spreading fire.

Figure 1 — Original laboratory-hazard and emergency-priority guide



Safety clarification: Never use a domestic refrigerator for flammable liquids; only specifically rated laboratory storage is suitable. Do not discharge chemical waste to a drain unless the institution's approved procedure and local rules explicitly permit it. A suspected explosive or unknown spill is an evacuation/isolation task, not a student cleanup task.

A. Multiple Choice Questions

1. Safety in the chemistry laboratory is:

(a) the responsibility of students only
(b) the responsibility of the professor only
(c) the responsibility of the lab incharge only
(d) a shared responsibility

Answer: (d) a shared responsibility — Students, teachers, technicians and management all have duties: follow rules, provide training/supervision, maintain equipment and stop unsafe work.

2. Accidents often result from:

(a) making mistakes
(b) failure to use common sense
(c) failure to follow instructions
(d) all of the above

Answer: (d) all of the above — Unsafe acts, poor judgement and ignored instructions can each contribute; planning and supervision reduce their likelihood and consequences.

3. The label ‘Warning’ on a chemical bottle signifies:

(a) that the chemical can cause less serious injury
(b) that the chemical can cause serious injury
(c) that the user should be careful when using the chemical
(d) that users should open it only in the presence of a teacher

Answer: (c) that the user should be careful when using the chemical — The intended answer is (c): read the label/SDS and take precautions. In modern GHS wording, ‘Warning’ is a signal word for lower-severity categories than ‘Danger’, not a statement that the material is safe.

4. The label ‘Corrosive’ on a chemical bottle indicates:

(a) that the material is an oxidizing agent
(b) that the material can degrade rapidly upon exposure
(c) that its contact destroys living tissue
(d) that the chemical can explode

Answer: (c) that its contact destroys living tissue — Corrosives can severely damage skin, eyes and other tissue; some also attack metals. The label does not by itself mean oxidizing or explosive.

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5. Example of a highly toxic chemical:

(a) ethanol (b) acetic acid
(c) potassium cyanide (d) potassium permanganate

Answer: (c) potassium cyanide — Potassium cyanide is acutely poisonous because cyanide interferes with cellular respiration. The other substances can also be hazardous, but are not the best answer.

6. Example of a self-reactive chemical:

(a) potassium (b) phenol
(c) picric acid (d) n-hexane

Answer: (c) picric acid — Picric acid is the book's example of a shock-/heat-sensitive energetic compound, especially when dried or improperly stored. It must not be handled casually.

7. When diluting an acid with water:

(a) do it quickly (b) do not stir the container
(c) always add acid to water (d) always add water to acid

Answer: (c) always add acid to water — Add acid slowly to a larger volume of water with suitable cooling/stirring as directed. Adding water to concentrated acid can cause violent local boiling and splashing.

8. What should you do in case of a fire drill in the lab?

(a) run to the safety shower (b) climb into the fume cupboard
(c) close gas valves and turn off all equipment if safe (d) carry chemicals out of the lab

Answer: (c) close gas valves and turn off all equipment if safe — Make the area safe only if this does not delay evacuation, then follow the drill route and assembly procedure. Never hide in a fume cupboard or carry chemicals out.

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B. Questions for Short Answers

13.1. Name some corrosive chemicals.

Examples include hydrochloric acid (HCl), sulphuric acid (H_2SO_4), nitric acid (HNO_3), hydrofluoric acid (HF), sodium hydroxide (NaOH), potassium hydroxide (KOH), glacial acetic acid and bromine. Concentration affects severity. HF requires substance-specific emergency arrangements because exposure can cause deep and systemic injury.

13.2. What safety precautions avoid damage due to explosive chemicals?

• Use prior risk assessment, trained supervision, the smallest feasible quantity and a safer substitute when possible.

• Avoid heat, flames, friction, impact, contamination and incompatible storage; inspect containers and dates as the SDS requires.

• Use suitable shielding, ventilation and remote/specialized procedures where required; keep people clear.

• **Do not move, open or clean a suspect container** (for example, dried material around a closure). Isolate the area and call trained emergency personnel.

13.3. What type of damage can reactive chemicals cause?

Uncontrolled reaction can release heat, flames, toxic or corrosive gases and high pressure, causing burns, respiratory injury, blindness, poisoning, explosions and damage to apparatus or buildings. Water-reactive metals/hydrides, oxidizers, azides and peroxide-forming materials require substance-specific segregation and handling.

13.4. Indicate two safety instructions required to avoid radiation.

Any two valid controls include:

1. Shield the source appropriately and keep it in labelled, controlled storage; shielding material depends on radiation type.

2. Minimize time and maximize distance from the source; use trained operation and access control.

3. Monitor exposure with suitable dosimeters/survey instruments and prevent ingestion, inhalation or skin contamination.

Follow the radiation-safety officer’s procedure; ordinary lab gloves or a mask do not stop penetrating gamma radiation.

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13.5. Which chemicals can cause suffocation?

Simple asphyxiants such as nitrogen, argon, helium and methane displace oxygen without necessarily having a warning odour; high CO_2 concentrations are also dangerous. **Chemical asphyxiants** such as carbon monoxide and hydrogen cyanide interfere with oxygen transport or cellular use. Toxic gases such as chlorine can additionally cause fatal respiratory injury.

13.6. Why are signs and symbols posted in the lab and on chemical bottles?

They communicate hazards and required action quickly, including to people who may not read long instructions. They support correct PPE, storage, segregation, waste handling and emergency response. A pictogram is only a starting point: read the full label and SDS because one container may have several hazards.

13.7. How should fire caused by chemicals be handled?

Raise the alarm, stop the experiment and isolate gas/power only if safe, then evacuate according to the emergency plan. A trained person may tackle a **small, incipient fire** with the correct extinguisher while keeping an escape route; otherwise leave and close doors if safe.

Use the SDS/fire plan to choose media: water can spread some burning liquids and is dangerous on energized equipment or water-reactive metals. Do not improvise. For burning clothing, use the safety shower or stop–drop–roll procedure and obtain medical aid.

13.8. Why are emergency drills important?

Drills turn written instructions into practised action. They teach alarm recognition, shutdown limits, evacuation routes, assembly/accounting, eyewash/shower locations and communication without panic. They also reveal blocked exits, missing equipment or unclear duties so these can be corrected before a real emergency.

C. Constructed Response Questions

13.1. How will you handle an emergency caused by fire due to short-circuiting?

1. Raise the alarm and keep clear. Do not touch the equipment, person or conducting surface while power may remain on.

2. Isolate electricity at a safe, accessible switch/breaker only if trained and this does not expose you to fire or shock.

3. Evacuate and call emergency help. Close doors if safe and provide the location and electrical-fire information.

4. Only a trained person with an escape route should use a suitable CO_2 or approved dry-powder extinguisher on a small electrical fire. Never use water on energized equipment.

Afterwards, keep the circuit isolated until a qualified electrician inspects it; do not reset the breaker repeatedly.

13.2. What type of reactions should be carried out in a fume cupboard?

Use a functioning fume cupboard for reactions that may release toxic, corrosive, irritating or flammable vapours/gases, volatile solvents, aerosols or hazardous dusts—subject to the risk assessment and hood design.

Examples include heating volatile acids/solvents and reactions generating Cl_2, SO_2, NO_x, H_2S or NH_3. Keep the sash at the marked height and work behind it. A normal fume hood is not a storage cupboard or an all-purpose blast shield; perchloric-acid and certain explosive operations require specialized equipment.

13.3. Put forward at least two suggestions to improve safety in the lab.

• **Assess and supervise:** written risk assessments, trained instructors, current SDS access and no unsupervised student work.

• **Control exposure:** maintain fume hoods, use minimum quantities and enforce suitable goggles, lab coat, gloves and closed footwear.

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• **Label and segregate:** compatible chemical storage, clear waste containers and secure gas cylinders.

• **Maintain and practise:** inspect electrical/gas equipment, keep exits/eyewash/showers clear and hold documented emergency drills.

The best programme combines engineering controls and safe procedures; PPE alone is not sufficient.

13.4. Can you identify warning symbols for radiation and asphyxiant chemicals?

Ionizing radiation: the international **trefoil**, commonly black on yellow (or magenta/black on yellow in some standards). It indicates a radiation area/source and requires access control and specified shielding/monitoring.

Asphyxiation: there is not one universal symbol for every asphyxiant. Compressed/liquefied gases often carry the GHS **gas-cylinder pictogram**, with supplementary “asphyxiant” wording; CO or HCN may instead/additionally show acute-toxicity or health-hazard pictograms. Always read the signal word, hazard statements and SDS.

An oxygen-deficiency monitor and ventilation are engineering controls; a pictogram alone cannot warn of an odourless oxygen-deficient atmosphere.

13.5. Why can sudden shock cause some chemicals to explode?

Shock or impact can supply enough activation energy to start rapid exothermic decomposition in an unstable energetic compound. Heat and a large volume of gas form in a very short time, producing a destructive pressure wave. Dry picric acid, some peroxides, azides, nitrocellulose and benzoyl peroxide illustrate why contamination, drying, friction and impact must be controlled by specialists.

D. Descriptive Questions

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13.1. Explain hazards due to explosive and toxic chemicals.

Hazard	How harm occurs	Examples / essential controls
Explosive / self-reactive	Rapid heat/gas/pressure release causes blast, fragments, burns, eye injury and property damage; shock, heat or contamination may initiate it.	Picric acid, azides, peroxides, nitrocellulose; minimum quantity, shielding, compatible storage, specialist response.
Acutely toxic	Small exposure can poison through inhalation, skin contact, ingestion or injection; effects may be immediate or delayed.	Cyanides, mercury compounds, chlorine and some pesticides; substitution, closed handling/fume hood, PPE and exposure monitoring.
Chronically toxic / carcinogenic	Repeated low exposures can damage organs, reproduction or development, or increase cancer risk.	Benzene and some heavy-metal compounds; avoid exposure, use occupational limits and health surveillance where required.

Toxic gas can also cause panic and secondary injury during evacuation. Emergency plans must address both immediate casualties and contamination of people, surfaces and air.

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13.2. Write five common safety instructions used to avoid all types of hazards.

1. Prepare: Read the method/SDS, complete a risk assessment, use minimum quantities and work only with authorization and supervision.

2. Protect: Wear splash goggles, lab coat, suitable gloves and closed footwear; add face shield or other task-specific PPE when required.

3. Control the workspace: Use ventilation/fume hood, keep exits clear, avoid flames/food/drinks and never taste or directly smell chemicals.

4. Label, segregate and dispose correctly: Keep incompatible chemicals apart, secure cylinders and use labelled waste streams—never unapproved drain disposal.

5. Be emergency ready: Know alarms, exits, eyewash/shower and extinguisher locations; report spills/exposure promptly and practise drills.

13.3. Explain the importance of warning signs and symbols in avoiding laboratory accidents.

Signs identify a hazard before contact and prompt a specific preventive action: eye protection, restricted entry, no ignition, shielding, ventilation or use of emergency equipment. Standard pictograms communicate quickly across language and literacy barriers and help workers select compatible storage, PPE and waste routes.

Emergency signs locate exits, showers, eyewash, first aid and fire equipment; prohibition signs prevent unsafe behaviour. To remain effective, signs must be legible, current, unobstructed and supported by training. A symbol does not replace the container label, SDS, risk assessment or supervision.

13.4. Name some toxic chemicals. Describe the effects of spreading toxic gas in the lab.

Examples include **chlorine, ammonia, hydrogen cyanide, carbon monoxide, mercury vapour, benzene, sulphur dioxide, nitrogen dioxide and hydrogen sulphide**. Their target organs and exposure limits differ.

A released gas may spread through the room and ventilation, causing coughing, eye/airway burns, headache, dizziness, nausea, unconsciousness or death. CO and HCN interfere with oxygen use; chlorine and NO_2 injure the lungs, sometimes with delayed worsening. Odour is not a reliable exposure test, and high H_2S can deaden smell.

Response: alert others, evacuate by the planned safe route, isolate the area and call trained responders. Do not enter for rescue without appropriate breathing apparatus and training. Provide the product name/SDS and obtain urgent medical assessment even if symptoms initially improve.

13.5. A student spills a corrosive and explosive chemical in an accident. Which emergency measures should be taken?

1. Do not touch, move, neutralize or clean it. Warn others, stop nearby work only if safe and evacuate immediately; avoid ignition, vibration and contamination.

2. Isolate the area and alert the instructor/emergency coordinator. Call specialist emergency services, give the chemical name/quantity if known and make the SDS available.

3. If skin or eyes are exposed, begin clean-water flushing immediately at the eyewash/shower while evacuation help is arranged; remove contaminated clothing where safe and obtain urgent medical aid. Do not use chemical neutralizers on a person.

4. Only trained responders should approach with substance-specific protective equipment, monitoring and an explosive/corrosive spill plan. Keep everyone else at the assembly point and account for them.

Afterwards, record injuries and exposure, arrange medical follow-up and investigate the cause before reopening the area.

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E. Investigative Question

13.1. A chlorine tanker leak spread over a large area in Lahore, killing people and animals. Give concrete proposals to prevent a future accident.

1. Prevent failure: Require certified tank/valve design, protective fittings, compatible materials, inspection/testing, maintenance records and safe filling limits.

2. Control transport: Licensed carriers, trained drivers, approved routes and times, secure parking, speed/fatigue control, vehicle tracking and prohibition of unsafe repairs.

3. Prepare the route: Shared shipping/SDS information, route-specific risk assessment, communications, emergency contacts and coordination with district/fire/health authorities.

4. Detect and respond early: Suitable leak-detection and trained responder teams with chlorine-compatible equipment, self-contained breathing apparatus, isolation distances and rehearsed casualty/decontamination plans.

5. Protect the public: Clear warning systems and drills for nearby communities. Follow official instructions to evacuate upwind/crosswind or shelter indoors as directed; avoid low areas because chlorine is denser than air.

6. Learn and enforce: Report and investigate incidents/near misses, publish corrective actions, audit contractors and enforce routing and maintenance standards.

Untrained people must never approach a chlorine cloud or attempt valve repair. Public warning should avoid sending residents through the contaminated plume.

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Points to Remember

Safety is shared: institution, instructor, technician and student each have responsibilities.

Acid dilution: add acid slowly to water according to the approved method; never add water to concentrated acid.

PPE complements—not replaces—substitution, ventilation, minimum quantities, supervision and safe design.

Radiation: time–distance–shielding plus access control and monitoring; ordinary PPE does not stop gamma rays.

Asphyxiants: N_2, Ar, He and CH_4 can displace oxygen; CO and HCN interfere with oxygen use.

Suspected explosive: do not move/open/clean; isolate, evacuate and call specialists.

Electrical fire: isolate power only if safe; no water on energized equipment; trained response only.

Toxic gas: evacuate and obtain trained help—odour is unreliable and unprotected rescue can create more casualties.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	d	2	d	3	c	4	c	5	c
6	c	7	c	8	c				

How to attempt this chapter in the paper

• For hazard questions, name the harm, one example and a matched control; “wear PPE” alone is not a complete answer.
• For emergencies, order the response: alert → evacuate/isolate → call trained help → substance-specific first aid/response.
• Distinguish corrosive, toxic, reactive, flammable and asphyxiant hazards; one chemical may belong to several classes.
• Use the SDS and institutional/local rules for disposal and spill response rather than assuming drain disposal is acceptable.

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